Fundamentals of Big Data - OEAD0001

Project title: RF Signal Strength Analysis and Coverage Visualization

Wireless communication networks generate RF signal data that can vary significantly across geographical locations due to distance, interference, network conditions, and environmental factors. However, raw RF measurements containing parameters such as signal strength, latitude, longitude, throughput, latency, and network type are difficult to interpret directly.

The problem is to develop a data-driven RF Signal Strength Analysis and Coverage Visualization system that processes RF measurement data, analyses the relationship between signal and network parameters, identifies signal-strength patterns across geographical regions, and represents the results through statistical visualizations and an interactive geographical coverage map.

The system will help users understand where RF signal strength is strong, moderate, or weak, while also providing insights into associated network performance parameters such as throughput and latency.

Step 1: Importing Required Libraries

Objective: To import the Python libraries required for data processing, numerical computation, statistical analysis, and visualization of the RF signal dataset.

1. **Pandas (pandas)** – Used for loading, cleaning, and manipulating the RF dataset.
2. **NumPy (numpy)** – Used for numerical operations and array-based calculations.
3. **Matplotlib (matplotlib.pyplot)** – Used to create graphs and visualize RF signal characteristics.
4. **Seaborn (seaborn)** – Used for statistical visualizations such as distributions and correlation matrices.
5. **%matplotlib inline** – Enables plots to be displayed directly inside the Jupyter Notebook.

```
Libraries imported successfully!
```

Step 2: Loading the Raw Dataset

Objective: To load the raw RF signal measurement dataset into a Pandas Data Frame so that it can be examined and analyzed in the subsequent steps.

The pandas.read_csv() function is used to read signal_metrics.csv from the project's raw data directory.

```
Dataset loaded successfully!
Number of rows: 16829
Number of columns: 12
```

Dataset Link: Click here to view Dataset

Step 3: Previewing the Dataset

Objective: To inspect the first few records of the loaded RF dataset and understand how the data is structured before performing further analysis.

The df.head() function displays the first **5 rows** of the Data Frame by default.

	Timestamp	Locality	Latitude	Longitude	Signal Strength (dBm)	Signal Quality (%)	Data Throughput (Mbps)	Latency (ms)	Network Type	BB60C Measurement (dBm)	srsRAN Measurement (dBm)	BladeRFxA9 Measurement (dBm)
0	2023-05-05 12:50:40.000000	Anisabad	25.599109	85.137355	-84.274113	0.0	1.863890	129.122914	3G	0.000000	0.000000	0.000000
1	2023-05-05 12:53:47.210173	Fraser Road	25.433286	85.070053	-97.653121	0.0	5.132296	54.883606	4G	-95.810791	-105.452359	-99.920892
2	2023-05-05 12:56:54.420346	Boring Canal Road	25.498809	85.211371	-87.046134	0.0	1.176985	119.598286	LTE	-91.593861	-95.419482	-87.714070
3	2023-05-05 13:00:01.630519	Danapur	25.735138	85.208400	-94.143159	0.0	68.596932	46.598387	5G	-90.642773	-101.895905	-96.570698
4	2023-05-05 13:03:08.840692	Phulwari Sharif	25.538556	85.159860	-94.564765	0.0	38.292038	30.342828	5G	-90.489100	-103.318304	-95.102467

The first few records of the dataset were displayed successfully, confirming that the RF measurement data was loaded correctly and is ready for detailed analysis.

Step 4: Identifying Dataset Columns

Objective: To identify and list all the features/attributes available in the RF signal dataset.

The df.columns.tolist() function extracts the names of all columns from the Data Frame and displays them as a list.

Step 5: Checking Data Types and Non-Null Values

Objective: To examine the structure and data quality of the dataset by checking the data type of each column and the number of non-null values.

The df.info() function provides a summary of the Data Frame.

```
RangeIndex: 16829 entries, 0 to 16828
Data columns (total 12 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Timestamp                     16829 non-null  object
 1   Locality                      16829 non-null  object
 2   Latitude                      16829 non-null  float64
 3   Longitude                     16829 non-null  float64
 4   Signal Strength (dBm)         16829 non-null  float64
 5   Signal Quality (%)            16829 non-null  float64
 6   Data Throughput (Mbps)        16829 non-null  float64
 7   Latency (ms)                  16829 non-null  float64
 8   Network Type                  16829 non-null  object
 9   BB60C Measurement (dBm)       16829 non-null  float64
 10  srsRAN Measurement (dBm)      16829 non-null  float64
 11  BladeRFxA9 Measurement (dBm)  16829 non-null  float64
dtypes: float64(9), object(3)
memory usage: 1.5+ MB
```

The dataset structure and data types were successfully examined. The non-null counts provide an initial check for missing values, while the data types confirm whether the variables are appropriately represented for subsequent analysis.

Step 6: Checking for Missing Values

Objective: To identify whether the RF dataset contains any missing or null values that could affect the accuracy of further analysis.

The df.isnull().sum() function checks every column and counts the number of missing values present in each feature.

```
Timestamp                       0
Locality                        0
Latitude                        0
Longitude                       0
Signal Strength (dBm)           0
Signal Quality (%)              0
Data Throughput (Mbps)          0
Latency (ms)                    0
Network Type                    0
BB60C Measurement (dBm)         0
srsRAN Measurement (dBm)        0
BladeRFxA9 Measurement (dBm)    0
dtype: int64
```

Step 7: Checking for Duplicate Records

Objective: To identify whether the RF dataset contains any duplicate rows that could lead to repeated measurements and affect the accuracy of the analysis.

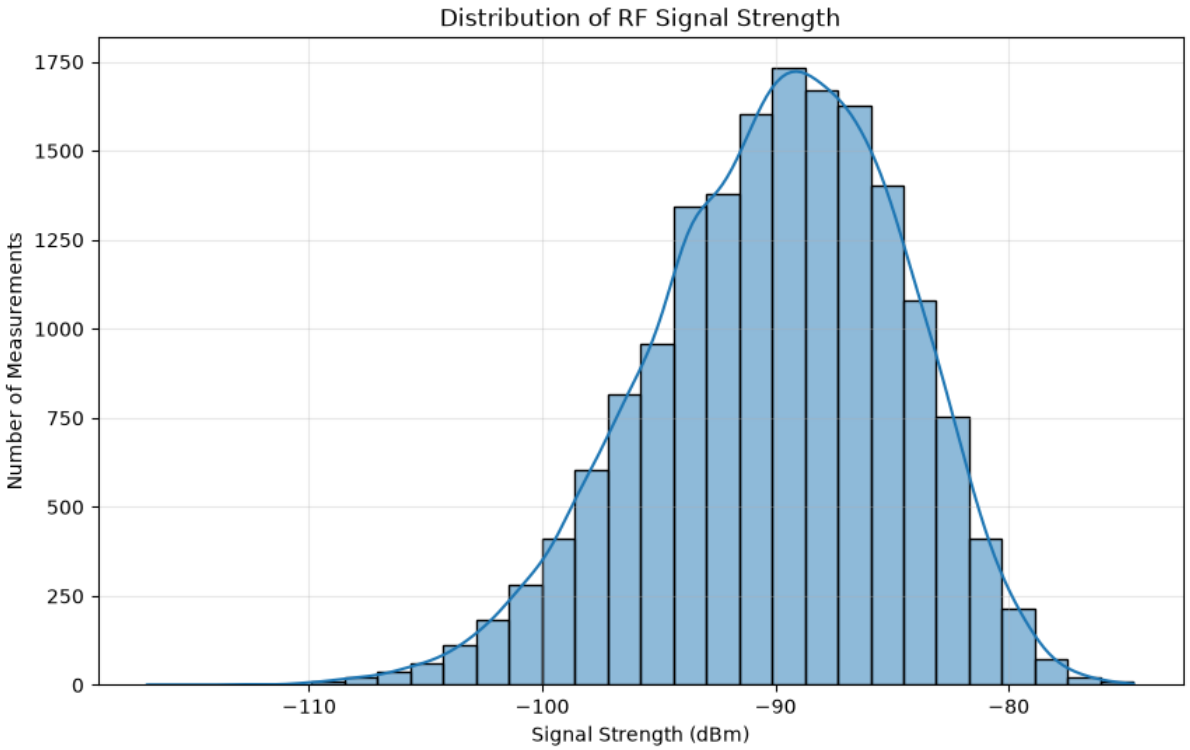
The df.duplicated().sum() function checks the complete dataset and counts the number of rows that are exact duplicates of previously occurring rows.

```
Duplicate rows: 0
```

Step 8: RF Signal Strength Distribution

Objective: To analyze how the RF signal strength values are distributed across all measurements in the dataset.

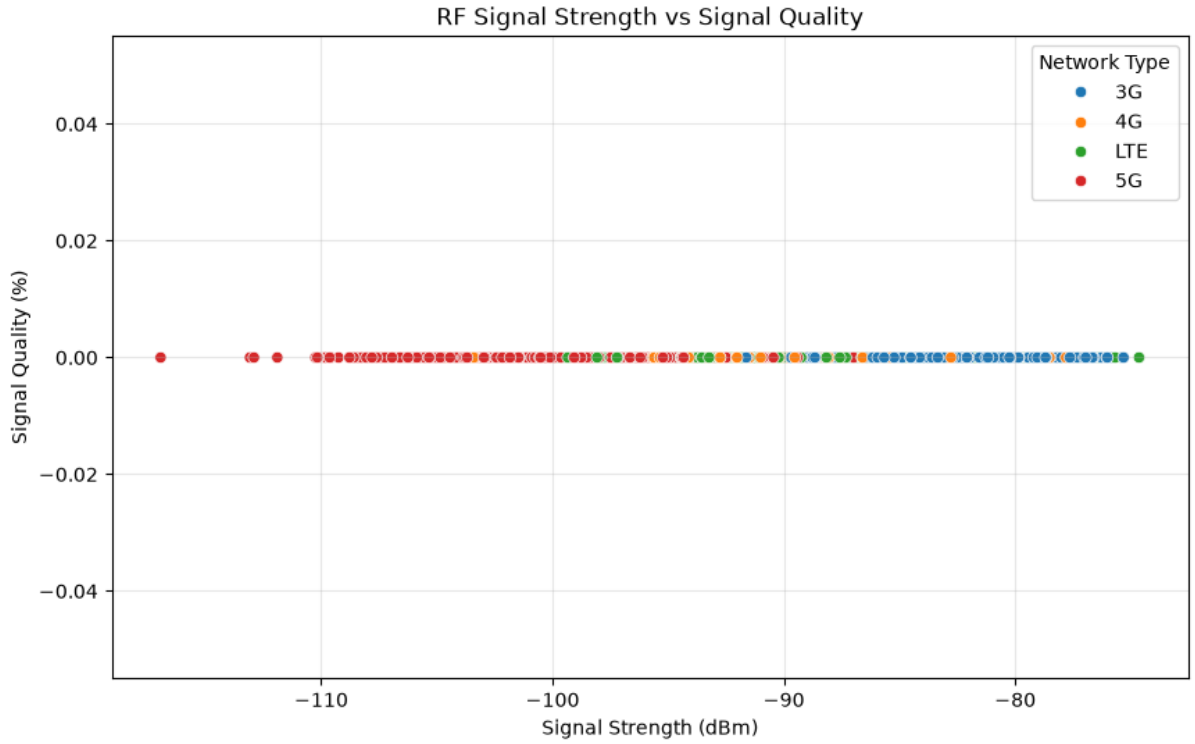
A histogram is created using the Signal Strength (dBm) column with 30 bins. The KDE (Kernel Density Estimate) curve is added to show the overall shape of the signal-strength distribution.



The plot shows that most RF measurements are concentrated approximately between −95 dBm and −85 dBm, with the highest concentration around −90 dBm. A smaller number of measurements occur at the weaker and stronger ends of the observed range.

Step 9: Signal Strength vs Signal Quality

Objective: To analyze the relationship between RF signal strength and signal quality and observe how signal quality varies at different signal-strength levels.



The scatter plot provides a visual comparison between signal strength and signal quality. It helps identify the general relationship between these two RF parameters and observe whether different network types show different patterns.

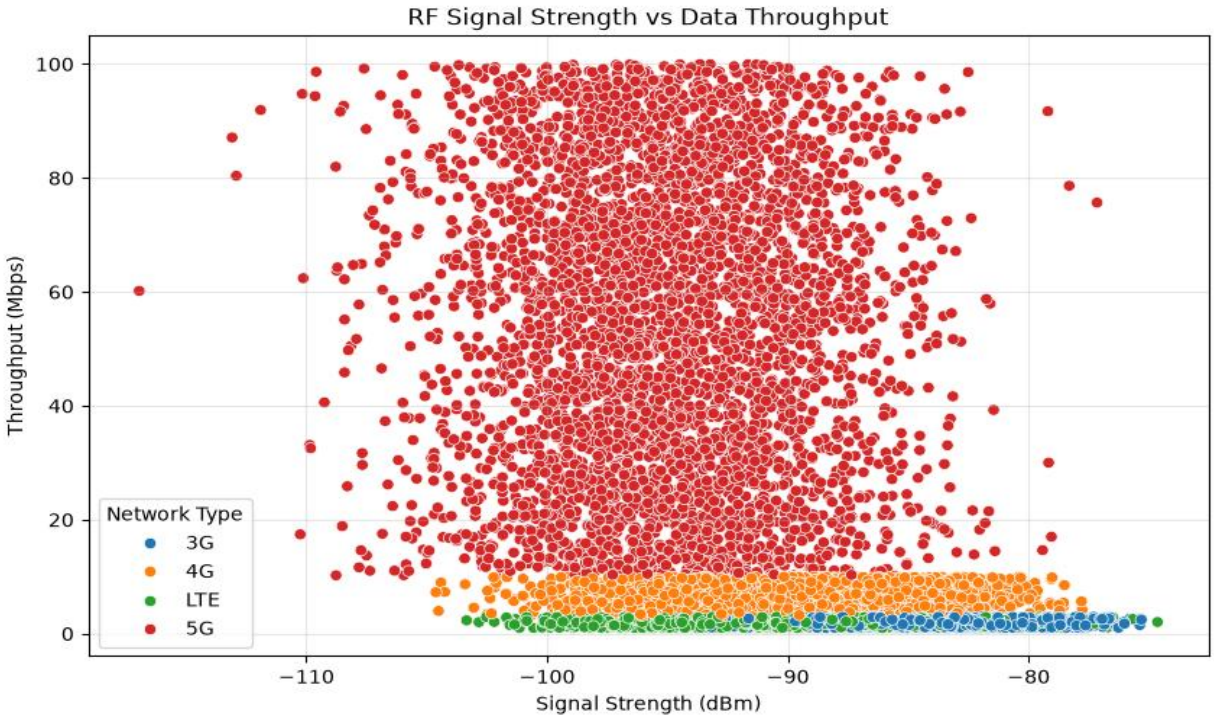
Step 10: Signal Strength vs Data Throughput

Objective: To examine the relationship between RF signal strength and data throughput and determine how network performance varies with different signal-strength levels.

A scatter plot is generated using:

- X-axis: Signal Strength (dBm)
- Y-axis: Data Throughput (Mbps)
- Colour grouping: Network Type

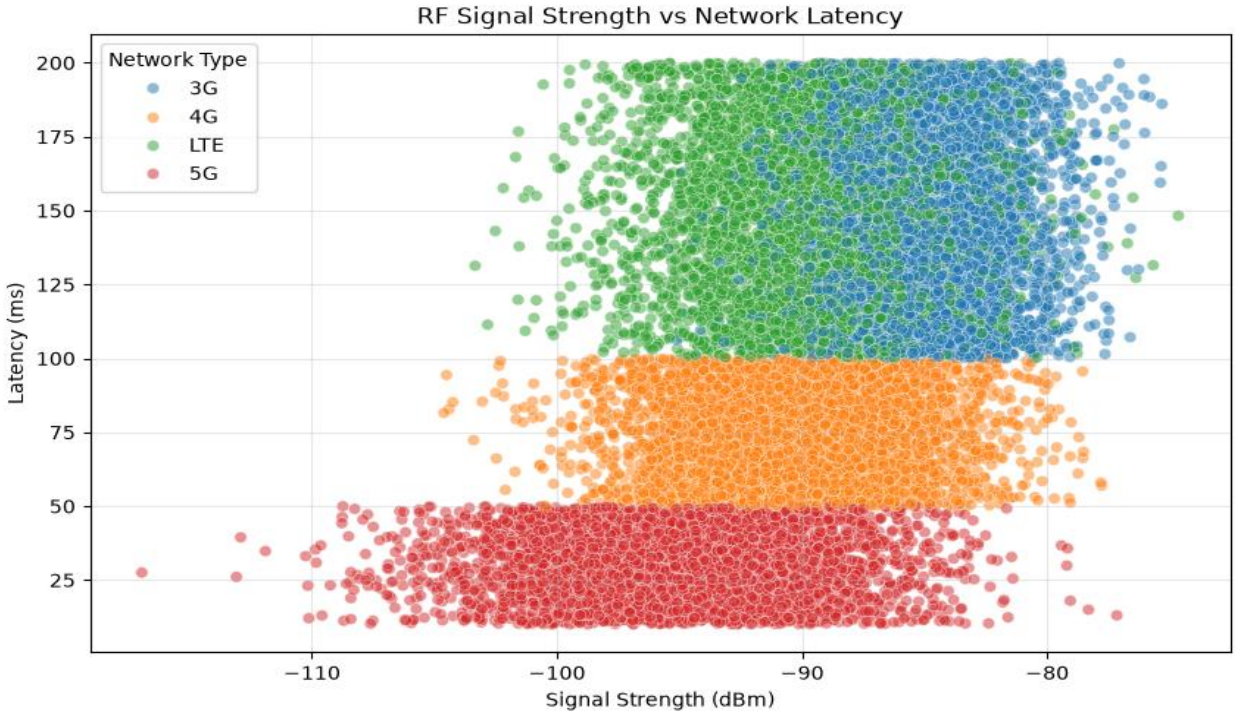
Each point represents an individual RF measurement, while the colour indicates the corresponding network type.



The visualization shows how data throughput varies across different RF signal-strength levels. It also allows comparison of throughput behavior among different network types.

Step 11: RF Signal Strength vs Network Latency

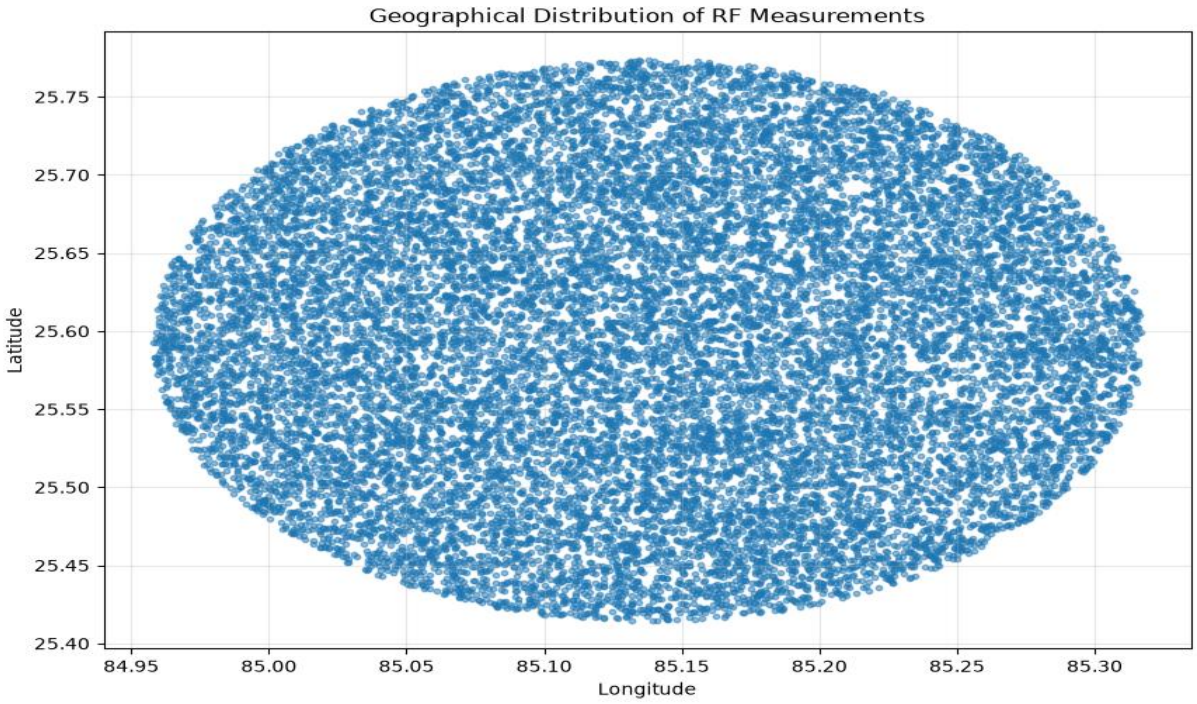
Objective: To analyze the relationship between RF signal strength and network latency and understand how communication delay varies with different signal conditions.



The visualization shows how network latency varies across different RF signal-strength levels and allows comparison between different network types.

Step 12: Geographical Distribution of RF Measurements

Objective: To visualize the spatial distribution of RF measurements based on their geographical coordinates and understand the area covered by the dataset.



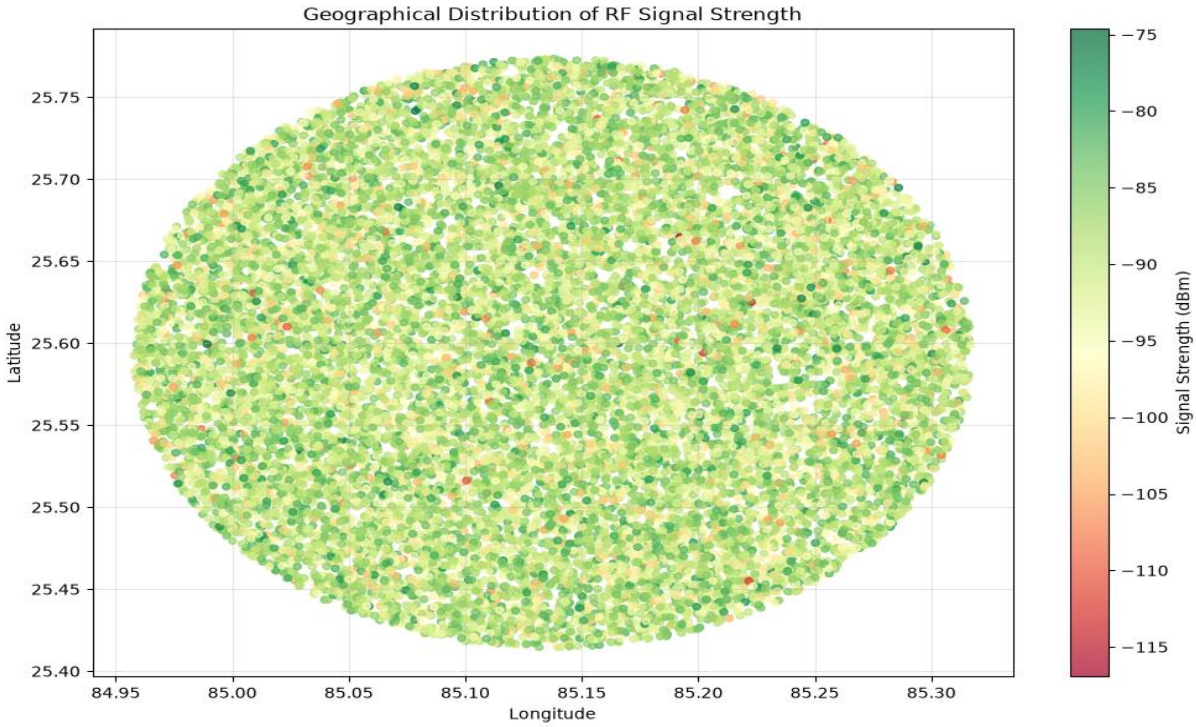
Each point corresponds to a recorded RF measurement at a specific geographical location. It confirms that the measurements are distributed across a defined geographical region rather than being concentrated at a single location.

Step 13: Geographical RF Signal Strength Visualization

Objective: To visualize how RF signal strength varies across different geographical locations using latitude and longitude coordinates.

A geographical scatter plot is created where:

- **X-axis**: Longitude
- **Y-axis**: Latitude
- **Point colour**: Signal Strength (dBm)
- **Colour map**: RdYlGn
- **Point size**: 20
- **Transparency**: 0.7



The plot provides a geographical view of RF signal-strength variation across the measurement area. Locations with different signal-strength values are represented using different colours, making spatial differences in RF conditions easier to identify.

Step 14: Statistical Analysis of Signal Strength

Objective: To calculate the key statistical measures of RF signal strength and understand the overall range and distribution of the measurements.

```
Minimum Signal Strength: -116.94226674903167
Maximum Signal Strength: -74.6448479734431
Mean Signal Strength: -90.07248449577469
Median Signal Strength: -89.66556554102601

Percentiles:
0.25   -93.615962
0.50   -89.665566
0.75   -86.145491
Name: Signal Strength (dBm), dtype: float64
```

The following statistical measures are calculated for Signal Strength (dBm):

- **Minimum** – The weakest recorded signal strength.
- **Maximum** – The strongest recorded signal strength.
- **Mean** – The average signal strength across all measurements.
- **Median** – The middle value when all signal-strength measurements are ordered.
- **25th percentile (Q1)** – Indicates the value below which 25% of measurements fall.

- **50th percentile (Q2)** – Equivalent to the median.
- **75th percentile (Q3)** – Indicates the value below which 75% of measurements fall.

The statistical values provide a numerical summary of the RF signal-strength distribution and help identify the typical signal level, overall range, and spread of the measurements.

Step 15.1: Classification of RF Signal Strength

Objective: To classify the measured RF signal strengths into three categories — Weak, Moderate, and Strong — based on the statistical distribution of the dataset.

The 25th percentile (Q1) and 75th percentile (Q3) obtained in Step 14 are used as thresholds.

The classification rule is:

- Weak: Signal strength ≤ Q1
- Moderate: Q1 < Signal strength ≤ Q3
- Strong: Signal strength > Q3

```
25th Percentile: -93.61596246906991
75th Percentile: -86.14549102908593

Signal Category Counts:
Signal Category
Moderate    8414
Weak        4208
Strong      4207
Name: count, dtype: int64
```

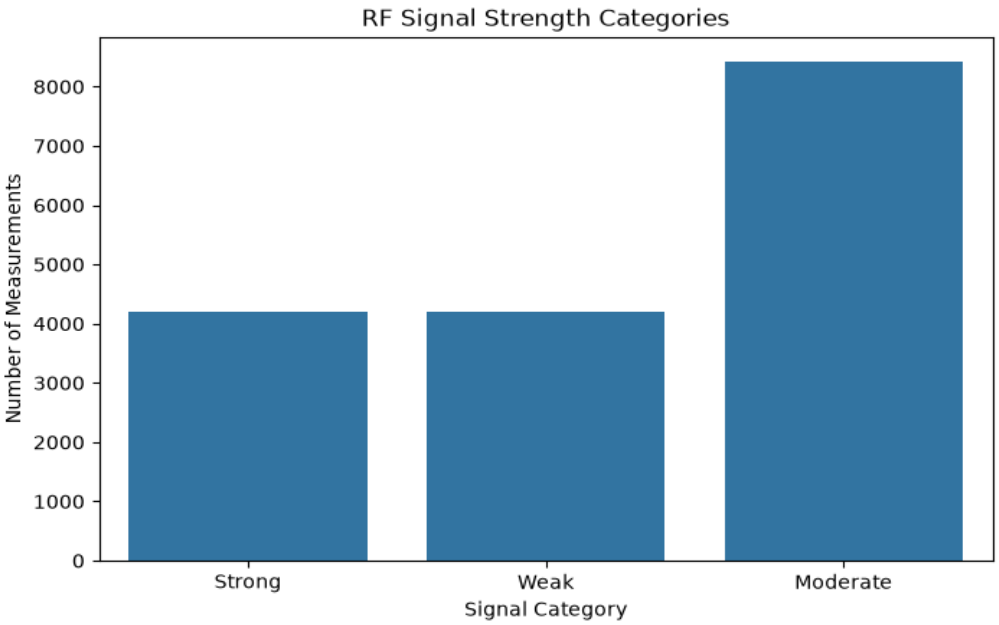
A new column called Signal Category is added to the dataset to store the classification for every RF measurement.

Step 15.2: Visualization of RF Signal Categories

Objective: To visualize the distribution of RF measurements across the three signal-strength categories: Weak, Moderate, and Strong.

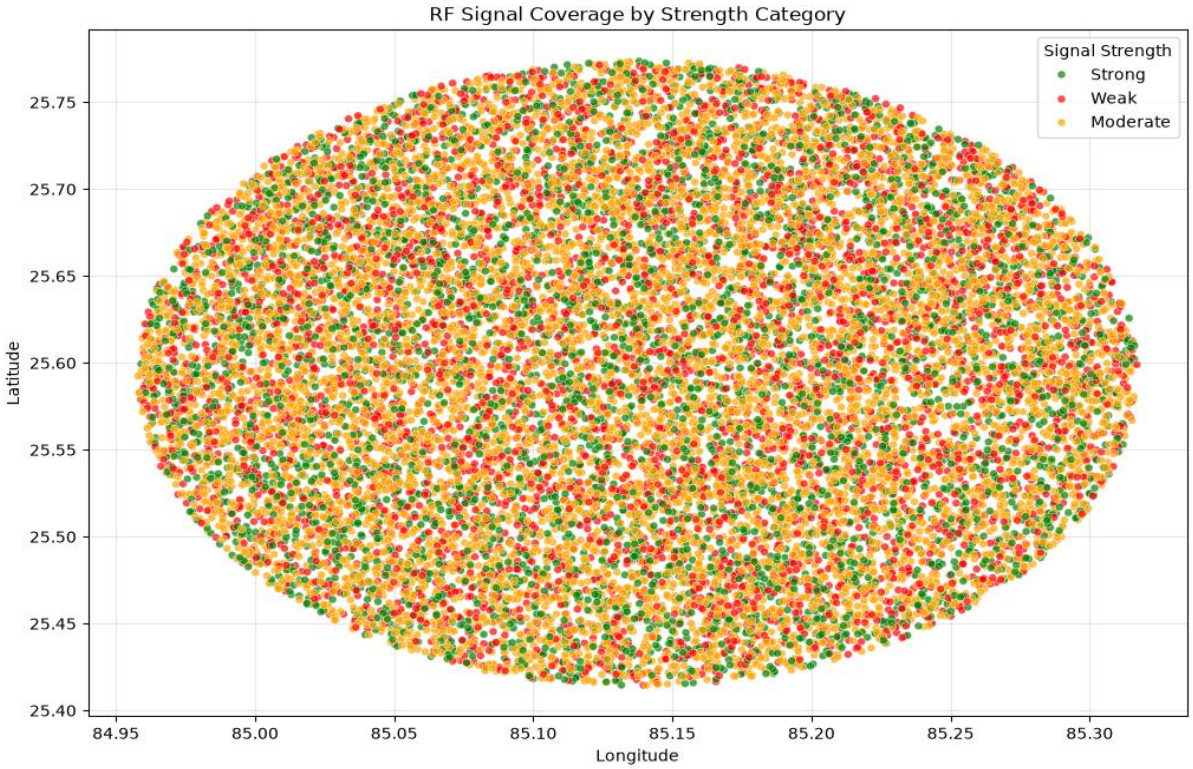
A count plot is created using the Signal Category column.

- **X-axis:** Signal Category
- **Y-axis:** Number of Measurements



Step 16: Geographical RF Coverage by Signal Category

Objective: To visualize the geographical distribution of RF signal categories and identify areas with Weak, Moderate, and Strong signal conditions.

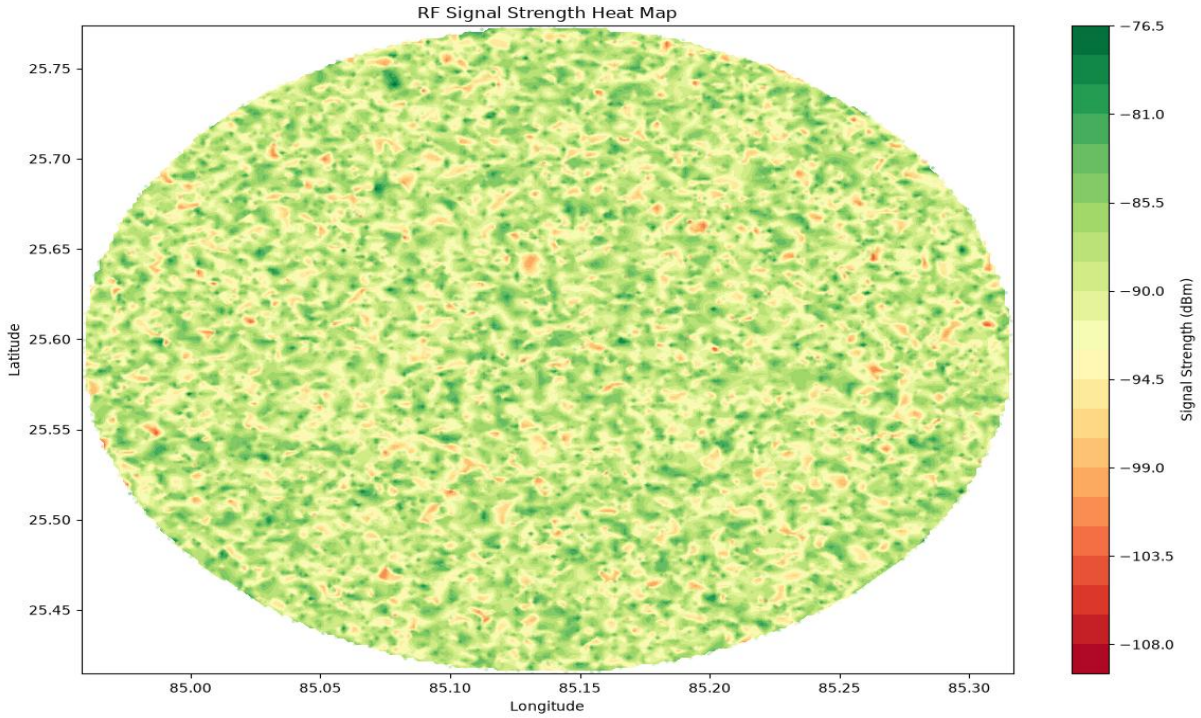


The visualization shows how Weak, Moderate, and Strong RF signals are distributed spatially across the measurement region.

- **X-axis:** Longitude
- **Y-axis:** Latitude
- **Red:** Weak signal
- **Orange:** Moderate signal
- **Green:** Strong signal

Step 17: Creating the RF Signal Strength Heat Map

Objective: To generate a continuous geographical heat map that estimates how RF signal strength varies across the entire measurement region, including locations where no direct measurement was recorded.



This step uses interpolation to transform individual RF measurements into a continuous geographical surface.

- Extract Latitude, Longitude, and Signal Strength directly from df.
- Create a 200 × 200 geographical grids covering the range of the available coordinates.
- Apply scipy.interpolate.griddata() using linear interpolation to estimate signal strength between the measured locations.
- Generate the heat map using contourf().
- Overlay the original measurement points using scatter().
- Add a colour bar to represent the corresponding Signal Strength (dBm).

The interpolation here is not creating new measured data. The values between measurement points are estimated from nearby observations using linear interpolation.

Step 18: Validation of the Interpolated Heat Map

Objective: To evaluate how accurately the interpolation method represents the original RF signal-strength measurements.

Step 17 creates the heat map, but we should also check whether the interpolation is reasonably consistent with the available measurements.

```
Heat Map Validation
-------------------
Valid measurements: 16829
Mean Absolute Error (MAE): 0.0 dBm
Root Mean Square Error (RMSE): 0.0 dBm
```

- **MAE (Mean Absolute Error)** – measures the average absolute difference between actual and interpolated values.
- **RMSE (Root Mean Square Error)** – measures the magnitude of the interpolation error while giving greater weight to larger errors.

This provides a quantitative validation of the interpolation process rather than relying only on visual interpretation of the heat map.

Step 19: Identifying Strongest and Weakest RF Locations

Objective: To identify the geographical locations with the strongest and weakest measured RF signal strengths in the dataset.

```
TOP 10 STRONGEST RF LOCATIONS
-----------------------------
       Locality  Latitude  Longitude  Signal Strength (dBm)
        Kumhrar 25.534256  85.028930             -74.644848
  Gandhi Maidan 25.599504  84.989190             -75.314531
    Pataliputra 25.599897  85.165807             -75.349049
Exhibition Road 25.741651  85.078505             -75.381195
 Rajendra Nagar 25.566981  85.121491             -75.672853
      Bankipore 25.649741  85.297338             -75.814159
    Fraser Road 25.696077  85.114397             -76.018573
Phulwari Sharif 25.628203  85.096641             -76.040852
        Danapur 25.625088  85.260116             -76.047085
    Pataliputra 25.515071  85.296669             -76.266148

TOP 10 WEAKEST RF LOCATIONS
---------------------------
         Locality  Latitude  Longitude  Signal Strength (dBm)
Patliputra Colony 25.664713  85.192150             -116.942267
       Kankarbagh 25.625183  85.222844             -113.082820
  Exhibition Road 25.594323  85.202219             -112.907955
      Bailey Road 25.455075  85.221732             -111.899047
       Kidwaipuri 25.630599  85.009329             -110.253857
          Danapur 25.516195  85.100526             -110.167665
       Kidwaipuri 25.525903  85.160490             -110.127088
      Boring Road 25.736879  85.156717             -109.858119
        Anandpuri 25.756864  85.192869             -109.824672
        S.K. Puri 25.610025  85.023490             -109.636501
```

The dataset is sorted based on the Signal Strength (dBm) column:

- nlargest(10) → identifies the 10 strongest RF measurements.
- nsmallest(10) → identifies the 10 weakest RF measurements.

Since dBm values are generally negative for RF measurements, values closer to 0 dBm represent stronger signals, while more negative values represent weaker signals.

Step 20.1: RF Signal Strength by Network Type

Objective: To compare the RF signal-strength characteristics across different network types present in the dataset.

The groupby() function groups the RF measurements according to **Network Type** and calculates the following statistics for each network:

```
RF Signal Strength by Network Type
-----------------------------------
              count   mean     min    max   std
Network Type
3G             4208 -85.01  -95.42 -75.31  3.03
4G             4219 -90.17 -104.62 -77.77  4.01
5G             4178 -95.15 -116.94 -77.16  4.95
LTE            4224 -90.00 -103.34 -74.64  3.97
```

- **Count** – Number of measurements recorded for that network type.
- **Mean** – Average RF signal strength.
- **Minimum** – Weakest recorded signal.
- **Maximum** – Strongest recorded signal.
- **Standard Deviation (std)** – Shows the variation of signal strength within that network type.

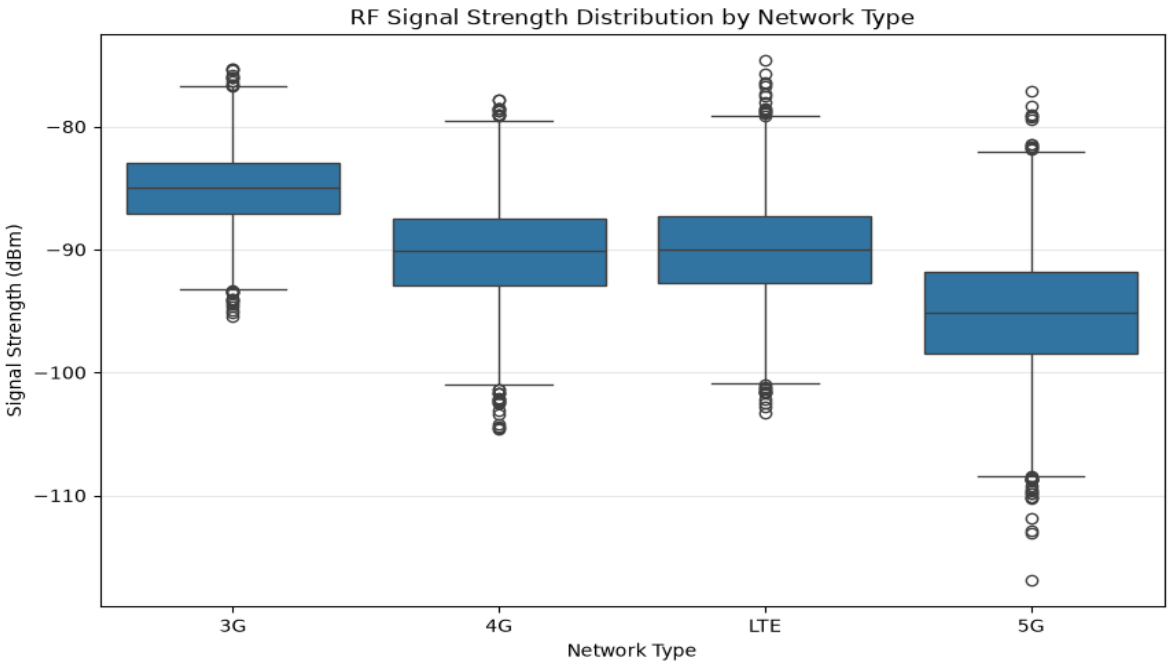
Step 20.2: RF Signal Strength Distribution by Network Type

Objective: To visually compare the distribution and variation of RF signal strength across different network types.

A box plot is created with:

- X-axis: Network Type
- Y-axis: Signal Strength (dBm)

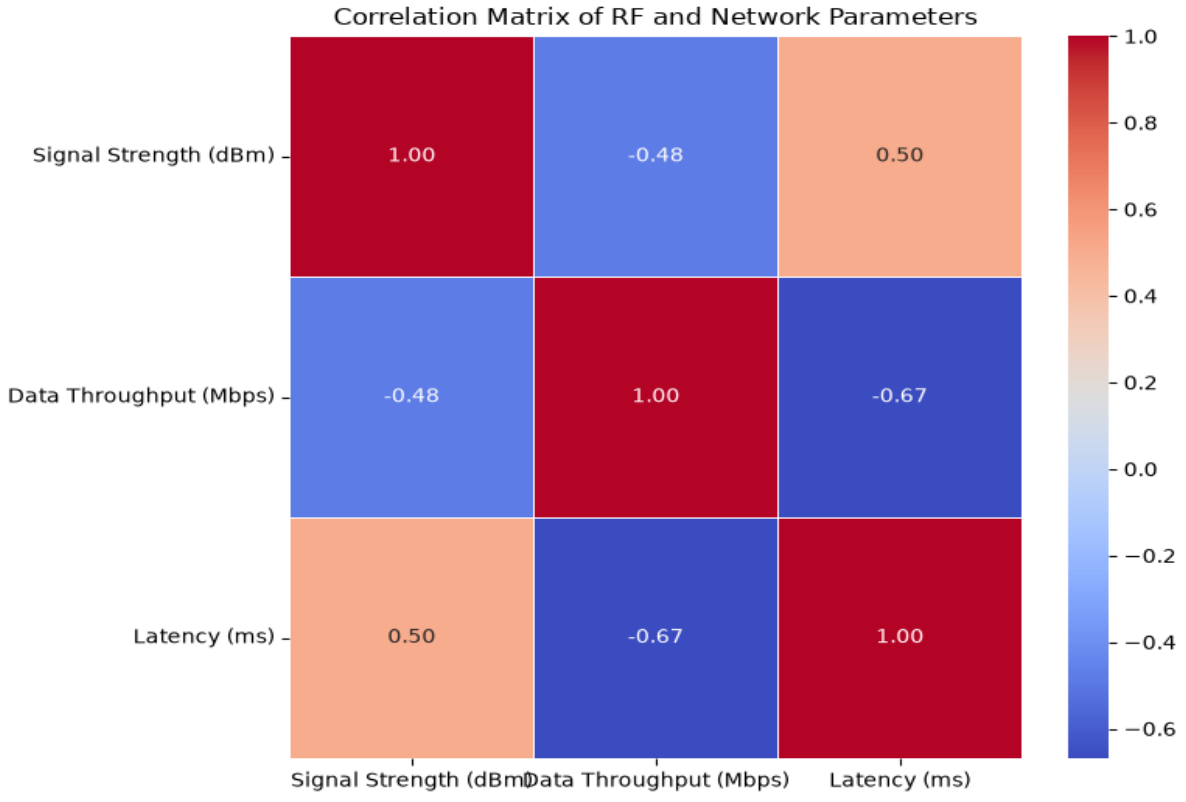
The box plot summarizes the signal-strength distribution for each network type using the median, quartiles, spread, and potential outliers.



The visualization makes it easier to compare the central signal strength, variability, and range among the different network types.

Step 21: Correlation Analysis of RF and Network Parameters

Objective: To analyze the linear relationships between RF signal strength and important network-performance parameters.



The .corr() function calculates the Pearson correlation coefficient between every pair of parameters.

The resulting matrix is represented as a heatmap, where each cell displays the correlation coefficient rounded to two decimal places.

The correlation coefficient ranges from **−1 to +1**:

- **+1** → Strong positive linear relationship
- **0** → Little or no linear relationship
- **−1** → Strong negative linear relationship

Step 22: Identifying Strongest and Weakest RF Regions

Objective: To identify geographical regions with consistently stronger or weaker RF signal conditions rather than evaluating individual measurement points.

TOP 10 STRONGEST RF REGIONS

Latitude	Longitude	Average Signal (dBm)	Measurements
25.50	84.98	-83.717538	3
25.74	85.08	-84.348069	9
25.64	84.96	-84.688474	3
25.41	85.13	-85.432328	1
25.44	85.04	-85.648557	2
25.57	85.12	-85.700839	13
25.54	85.02	-85.746013	15
25.67	85.22	-85.952821	11
25.69	85.12	-86.189574	16
25.42	85.19	-86.206901	6

TOP 10 WEAKEST RF REGIONS

Latitude	Longitude	Average Signal (dBm)	Measurements
25.75	85.04	-103.580589	1
25.76	85.21	-98.640485	3
25.69	84.98	-98.004196	1
25.75	85.23	-96.728359	3
25.77	85.08	-95.855523	1
25.41	85.14	-95.390245	2
25.77	85.09	-95.106078	2
25.52	85.10	-95.051234	10
25.60	85.29	-94.835770	17
25.54	85.08	-94.710342	15

Note: Step 19 is point-level analysis, while Step 22 is regional-level analysis.

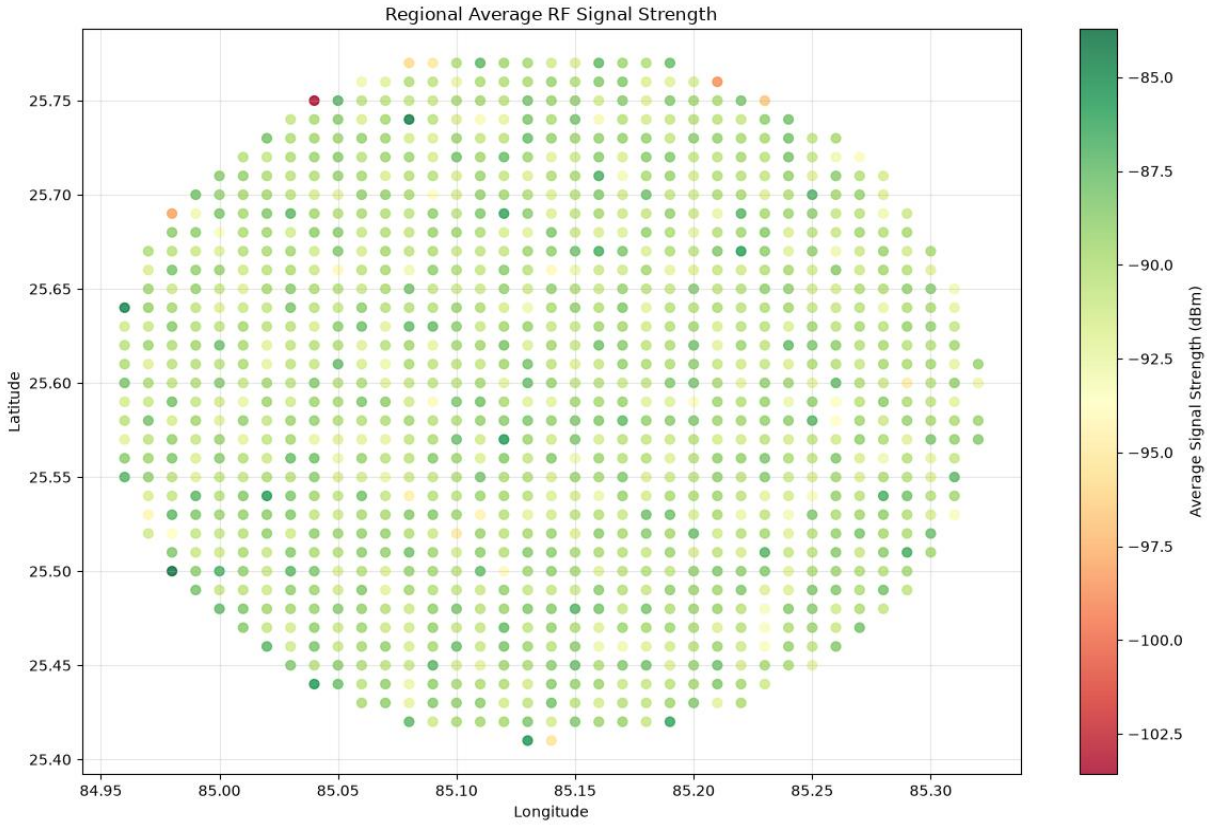
For each geographical region, the code calculates:

- **Average Signal (dBm)**: Mean signal strength of all measurements in that region.
- **Measurements**: Number of RF measurements recorded in that region.

Step 23: Regional Average RF Signal Map

Objective: To visualize the average RF signal strength across different geographical regions and identify spatial patterns in regional signal quality.

This step uses the region_signal Data Frame created in Step 22.



Each point now represents a geographical grid region, rather than an individual RF measurement. The colour bar indicates the average signal strength of each region.

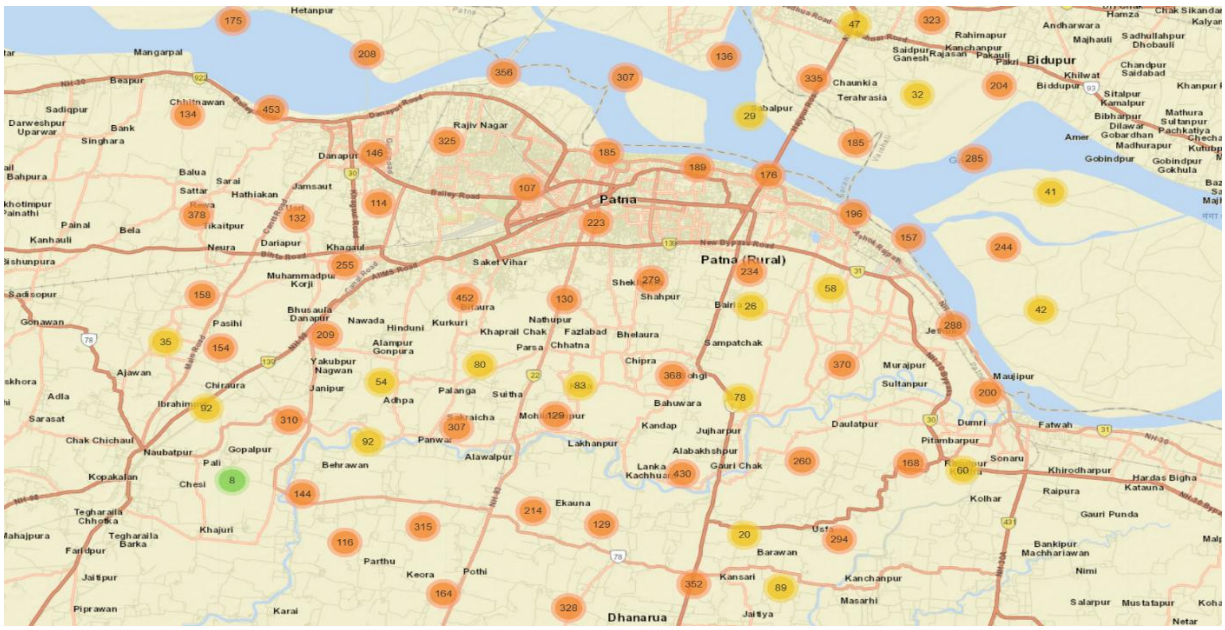
Step 24: Interactive RF Coverage Map

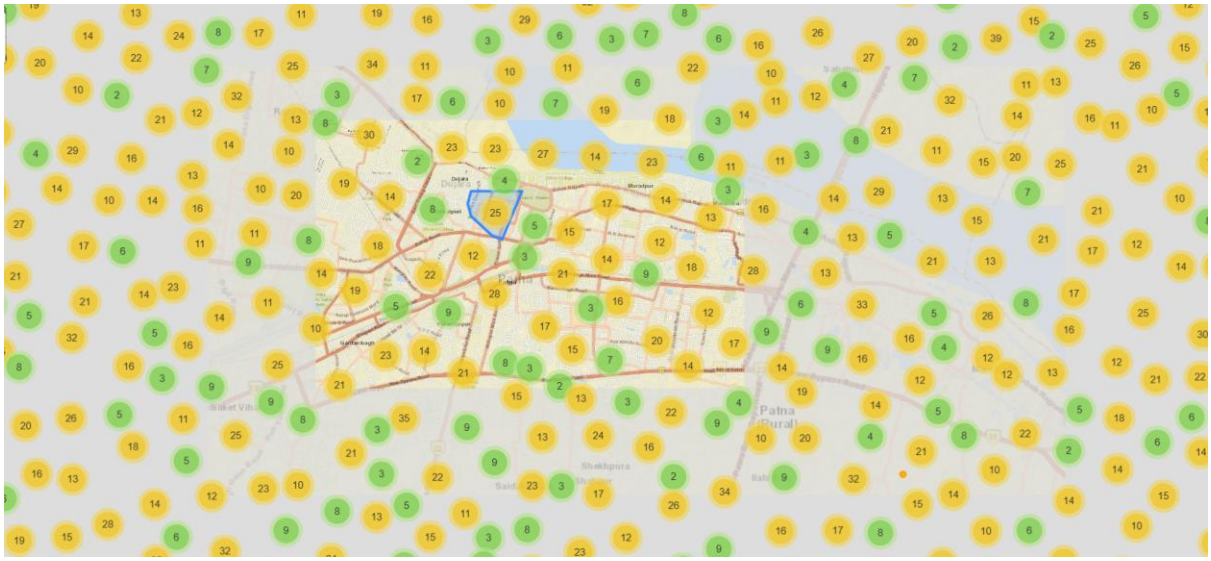
Objective: To create a real-world interactive geographical map that displays RF signal measurements at their actual locations and allows users to explore signal conditions interactively.

This step converts the previous analytical results into an interactive map using Folium.

1. The average latitude and longitude of all measurements are used as the initial map location.
2. A Folium map is created without a default tile layer.
3. An Esri World Street Map is added as the geographical base layer, allowing RF measurements to be viewed in the context of actual roads and geographical features.
4. ‘MarkerCluster’ groups nearby RF measurements when the map is zoomed out, reducing visual clutter when many measurements are close together.
5. Assign colours:
    - Strong: Greem
    - Moderate: Orange
    - Weak: Red
6. Every Measurement is represented using a ‘CircleMarker’ at its recorded latitude and longitude.









Zoom-Level Analysis: At lower zoom levels, nearby RF measurements are grouped into clusters to provide an overall view of the coverage. As the map is zoomed in, clusters split into smaller groups and individual measurement points become increasingly visible. This allows both regional RF coverage patterns and detailed measurement locations to be analyzed interactively.

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